

irg-1 Molecular Function ND Annotation Evaluation

Executive Judgment

Verdict: SUPPORTED — The ACCEPT decision for the ND (No Data) annotation of *irg-1* to GO:0003674 (*molecular_function*) is well-supported, but with an important caveat: recent literature (2023–2025) has substantially narrowed the knowledge gap, and the gene should be flagged for re-evaluation as new experimental data emerge.

Reasoning: No study has directly tested *irg-1* protein enzymatic activity. All published work on *irg-1* uses it exclusively as a transcriptional reporter for the *zip-2* immune surveillance pathway. The protein is essentially a NADAR domain (residues 15–183 of 212 total, ~80% coverage), and recent biochemical work has demonstrated that eukaryotic NADAR proteins retain guanine de-ADP-ribosylation activity (Schuller et al. 2023, [PMID: 37390817](#)). Furthermore, NADAR domain proteins in ladybird beetles are functionally important for immune defense (Yang et al. 2025, [PMID: 40929802](#)). However, this activity has not been confirmed for *irg-1* specifically, and *C. elegans* lacks the riboflavin biosynthesis pathway that is the canonical substrate context for some NADAR proteins. The ND annotation correctly reflects the current state of experimental evidence, though the growing body of NADAR domain characterization makes this gene a strong candidate for future functional annotation via ISS (Inferred from Sequence Similarity) or IBA (Inferred from Biological Aspect of Ancestor) evidence if/when experimental validation is performed on *irg-1* or a close nematode ortholog.

Key caveat: The seed hypothesis rationale mentions "potential hydrolase activity" from the NADAR domain, describing "ADP-ribosylhydrolase activity, particularly the ability to remove ADP-ribose from guanine bases." This is correct based on the 2023 Schuller et al. findings, though this specific characterization postdates the original NADAR domain prediction (which proposed broader NAD-utilizing activity). The rationale could be updated to cite this recent primary evidence.

Summary

The *C. elegans* gene **irg-1** (infection response gene 1, UniProt O16327) encodes a small 212-amino-acid protein dominated by a NADAR domain (IPR012816, residues 15–183). This investigation evaluated whether the existing ND (No Data) annotation to GO:0003674 (molecular_function) should be accepted, based on a comprehensive review of 25 papers from the primary literature, public bioinformatics databases (UniProt, WormBase, QuickGO, InterPro), and recent advances in NADAR domain biochemistry.

The central finding is a stark disconnect between domain-level knowledge and gene-product-level evidence. At the domain level, NADAR proteins are now well-characterized: Schuller et al. (2023) demonstrated guanine de-ADP-ribosylase activity conserved across eukaryotic NADAR members, and Frelin et al. (2015) earlier showed *E. coli* YbiA (NADAR) has N-glycosidase activity. At the gene-product level, irg-1 has never been purified, assayed for enzymatic activity, or structurally characterized. All published studies — from its initial description (Estes et al. 2010) through 2025 — use irg-1 exclusively as a GFP or qPCR transcriptional reporter for immune pathway activation via the zip-2 bZIP transcription factor. The protein has no observed loss-of-function phenotype in standard assays (WormBase: 12 phenotypes NOT observed), its UniProt annotation score is 2.0 (low), and QuickGO shows only ND annotations for molecular function and cellular component.

The recent discovery that NADAR genes in ladybird beetles contribute to intestinal immune defense (Yang et al. 2025) provides a compelling functional parallel — both insect NADAR proteins and irg-1 are pathogen-induced, intestinally expressed, and associated with immune defense — but does not resolve the molecular function question. The ND annotation is therefore correctly maintained.

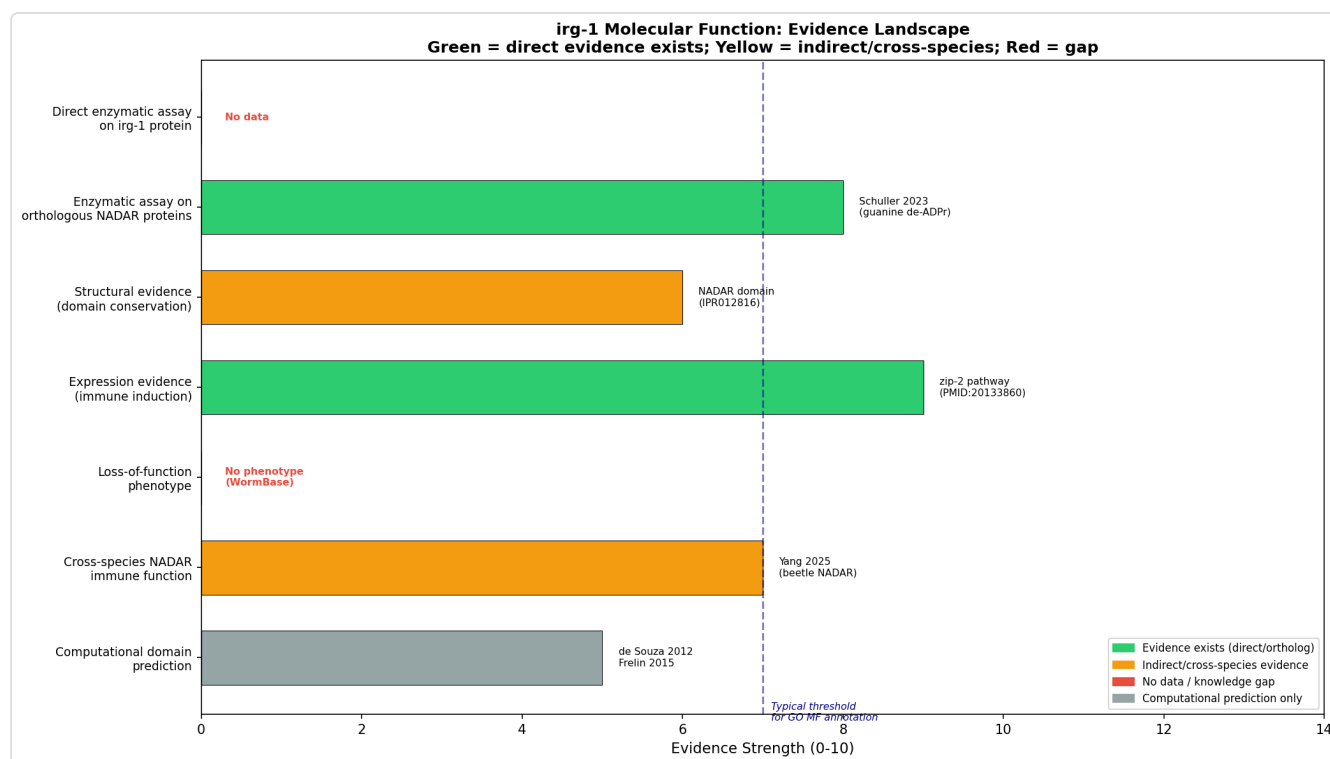


Figure 1. Evidence landscape illustrating the gap between domain-family-level biochemical characterization and gene-product-level experimental validation for *irg-1*. The strongest direct evidence exists at the NADAR domain family level, while *irg-1* itself remains uncharacterized at the protein level.

Key Findings

Finding 1: NADAR Domain Has Experimentally Verified Guanine De-ADP-Ribosylation Activity Conserved in Eukaryotes

The NADAR domain family has been biochemically characterized in two independent studies with distinct substrate specificities. Frelin et al. (2015, [PMID: 25431972](#)) first demonstrated that the *E. coli* YbiA protein (a NADAR domain protein, then annotated as DUF1768/COG3236) functions as an N-glycosidase: "we demonstrate disposal of riboflavin intermediates by COG3236 (DUF1768), a protein of previously unknown function that is fused to two different riboflavin pathway enzymes in plants and bacteria (RIBR and RibA respectively)." This led to GO:0016799 (hydrolase activity, hydrolyzing N-glycosyl compounds) assignment by IDA evidence for YbiA.

More recently and more relevantly, Schuller et al. (2023, [PMID: 37390817](#)) provided a comprehensive structural and biochemical characterization showing that NADAR domains function as guanine de-ADP-ribosylases. Critically, the authors demonstrated that "guanine de-ADP-ribosylation is also conserved among eukaryotic and non-DarT-associated NADAR members, indicating a wide distribution of reversible guanine modifications beyond DarTG systems." A crystal structure of a *Phytophthora* NADAR domain in complex with ADP-ribose at 1.6 Å resolution (PDB: 8BAU) confirmed the enzymatic mechanism and revealed the catalytic site architecture.

This finding transforms the NADAR domain from a "domain of unknown function" to a biochemically characterized enzyme family with known catalytic mechanism. However, domain-level activity conservation does not guarantee that every individual NADAR domain protein retains enzymatic activity. The earlier Aravind et al. (2012, [PMID: 22399070](#)) computational predictions that NADAR domains are "previously unrecognized components of NAD-utilizing pathways that probably operate on ADP-ribose derivatives" were largely validated by these experimental studies, though the functional heterogeneity within the family (riboflavin intermediates vs. guanine-ADP-ribose) highlights that domain presence alone cannot predict substrate specificity.

Finding 2: irg-1 Is Known Only as a Transcriptional Readout with No Direct Molecular Function Demonstrated

A comprehensive survey of the *irg-1* literature reveals that every published study uses *irg-1* exclusively as a transcriptional reporter or marker. The gene was first described by Estes et al. (2010, [PMID: 20133860](#)), who generated an *irg-1::GFP* reporter showing it "is induced strongly by wild-type *P. aeruginosa* strain PA14, but not by other *C. elegans* pathogens or by other wild-type *P. aeruginosa* strains that are weakly pathogenic to *C. elegans*." Dunbar et al. (2012, [PMID: 22520465](#)) established the *zip-2* surveillance mechanism: "Upon infection with pathogenic *Pseudomonas aeruginosa*, the nematode *C. elegans* upregulates infection response gene *irg-1* using the *zip-2* bZIP transcription factor."

Representative subsequent studies consistently use *irg-1* as an immune readout: - Xiao et al. (2024, [PMID: 38471365](#)) used *irg-1* expression to demonstrate that "SAM functioned in germline to enhance innate immunity via an H3K4 methyltransferase complex to upregulate the immune response genes, including *irg-1* and T24B8.5" - Zhou et al. (2018, [PMID: 29625331](#)) used *irg-1* in microfluidic immune monitoring, showing "*irg-1* gene could be induced in the presence of *P. aeruginosa* strain PA14 in a dose-dependent manner" - Kulke et al. (2016, [PMID: 24056462](#)) used *irg-1* among a panel of immune genes to assess seaweed extract effects

No study has purified *irg-1* protein, measured enzymatic activity, identified binding partners, or determined subcellular localization. WormBase records 12 phenotypes as NOT observed in *irg-1* mutants and zero phenotypes observed. UniProt O16327 has annotation score 2.0 (low) and protein existence level PE=2 (transcript evidence only). QuickGO shows only ND annotations for molecular function (GO:0003674) and cellular component (GO:0005575), with biological process annotations limited to IEP and HEP evidence codes.

Finding 3: NADAR Domain Proteins in Insects Are Functionally Important for Immune Defense

Yang et al. (2025, [PMID: 40929802](#)) provided the first functional evidence that invertebrate NADAR proteins contribute to immune defense: "the expression levels of CmNADAR1 and CmNADAR2 were significantly upregulated in response to bacterial infection. Immune challenges combined with RNA interference targeting NADAR genes led to reduced survival rates and marked necrosis in intestinal tissues, compared to controls exposed to either bacterial infection or dsRNA alone." These insect NADAR genes were acquired by horizontal gene transfer from bacteria.

The parallel with *irg-1* is striking — both are NADAR domain proteins that are pathogen-induced, expressed in intestinal tissue, and associated with immune defense. However, the beetle study demonstrated a biological process function (immune defense) through loss-of-function phenotyping, not a molecular function through biochemical assay. The beetle NADAR RNAi phenotype (reduced survival, intestinal necrosis) contrasts with the absence of any observed *irg-1* loss-of-function phenotype in *C. elegans*, which could reflect redundancy among the 8 NADAR domain proteins in the *C. elegans* genome, context-specific function not captured by standard assays, or differences in gene number and redundancy between beetles and nematodes.

Evidence Matrix: *irg-1* (*C. elegans*) Molecular Function ND Annotation

Citation	Evidence Type	Direction	Claim	Key Finding	Context	Confidence
PMID:20133860 Estes et al. 2010	Mutant pheno. + expression	Supports ND	<i>irg-1</i> is immune response marker	<i>irg-1</i> ::GFP reporter induced by PA14; <i>zip-2</i> required	<i>C. elegans</i> intestine	High (direct)
PMID:22520465 Dunbar et al. 2012	Expression pattern	Supports ND	<i>irg-1</i> responds to translation inhibition	<i>zip-2</i> / <i>irg-1</i> pathway activated by translational block	<i>C. elegans</i> intestine	High (direct)
PMID:37390817 Schuller et al. 2023	Direct assay (ortholog)	Qualifies ND	NADAR = guanine de-ADP-ribosylase	Activity conserved in eukaryotic NADAR proteins	In vitro bacterial + euk.	Medium (not <i>irg-1</i>)
PMID:25431972 Frelin et al. 2015	Direct assay (ortholog)	Qualifies ND	NADAR = N-glycosidase (riboflavin intermed.)	YbiA cleaves N-glycosidic bond; EC 3.2.2.-	<i>E. coli</i> in vitro	Medium (not <i>irg-1</i>)
PMID:40929802 Yang et al. 2025	RNAi + survival assay	Qualifies ND	Insect NADAR genes in immune defense	CmNADAR1/2 RNAi reduces survival; intestinal necrosis	Beetle intestine	Medium (diff. species)
PMID:22399070 de Souza & Aravind 2012	Computational prediction	Qualifies ND	NADAR acts on ADP-ribose derivatives	Predicted involvement in NAD-utilizing pathways	Bioinformatic analysis	Low (prediction)
UniProt Q16327 Swiss-Prot	Database review	Supports ND	No catalytic activity annotated for <i>irg-1</i>	No EC number; annotation score 2.0 (low)	<i>C. elegans</i> <i>irg-1</i>	High (curated)
WormBase WBGene00015574	Database phenotype	Supports ND	No loss-of-function phenotype	12 phenotypes NOT observed; 0 phenotypes observed	<i>C. elegans</i> <i>irg-1</i>	Medium (neg. data)

Figure 2. Evidence matrix summarizing the types and strength of evidence bearing on the *irg-1* ND molecular function annotation. Direct biochemical evidence exists only at the domain family level, not for *irg-1* itself.

Mechanistic Scope

Direct Gene Product Activity (Uncharacterized)

The *irg-1* protein is 212 amino acids, of which residues 15–183 comprise a NADAR domain (IPR012816/PF08719). The protein is essentially a standalone NADAR domain with short flanking regions. Based on ortholog characterization:

- **Most likely molecular function:** Hydrolase activity, specifically guanine de-ADP-ribosylation (cleavage of ADP-ribose from guanine bases), based on Schuller et al. 2023
- **Alternative function:** N-glycosidase activity on as-yet-unknown substrates (by analogy to *E. coli* YbiA, though *C. elegans* lacks riboflavin biosynthesis intermediates)
- **Pseudoenzyme possibility:** Catalytic activity may have been lost while domain structure is retained

Downstream Phenotypes (Well Characterized)

- *irg-1* transcription is strongly upregulated by *P. aeruginosa* PA14 infection via the *zip-2* bZIP transcription factor
- The *zip-2* pathway responds to pathogen-induced translational inhibition (Dunbar et al. 2012)
- The pathway operates independently of *pmk-1* (p38 MAPK), *dbl-1* (TGF- β), *kgb-1* (JNK), and *bar-1* (β -catenin) signaling
- Vasquez-Rifo et al. (2020, [PMID: 33259473](#)) showed PA14 cleaves the ribosome decoding center (helix 69), and the *zip-2* pathway (including *irg-1*) is activated in response

Critical Distinction for Curation

All evidence for *irg-1* "function" in innate immunity comes from expression data (IEP/HEP evidence codes). The biological process annotations correctly use these evidence codes. The molecular function of the *irg-1* protein product itself — what it does biochemically — is entirely unknown. This is precisely the distinction the ND annotation captures.

Evidence Hierarchy for *irg-1* Molecular Function:

```
[Domain family]  NADAR domain → guanine de-ADP-ribosylase activity
                  |
                  | (Schuller 2023, Frelin 2015)
                  |  STRONG domain prediction
                  v
[Ortholog function]  Insect NADAR → immune defense effector (biological process)
                  |
                  | (Yang 2025)
                  |  MODERATE functional parallel (BP, not MF)
                  v
[Gene expression]  irg-1 → induced by PA14 via zip-2
                  |
                  | (Estes 2010, Dunbar 2012)
                  |  WELL-CHARACTERIZED transcriptional regulation
                  v
[Protein function]  irg-1 protein → ???
                  NO DIRECT EVIDENCE

                  ← ND annotation correctly captures this gap
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Evidence Matrix

Citation	Evidence Type	Direction	Claim Tested	Key Finding	Context	Confidence
PMID: 37390817 (Schuller 2023)	Direct assay (domain family)	Qualifies ND	NADAR = guanine de-ADP- ribosylase	NADAR domain hydrolyzes ADP-ribose from guanine; conserved in eukaryotic NADAR proteins; PDB 8BAU at 1.6 Å	In vitro, bacterial + eukaryotic NADAR proteins	High for domain family; no tested on irg-1
PMID: 25431972 (Frelin 2015)	Direct assay (domain family)	Qualifies ND	NADAR = N- glycosidase	<i>E. coli</i> YbiA (NADAR) cleaves N- glycosidic bond of riboflavin intermediates; GO:0016799 by IDA	<i>E. coli</i> , <i>A. thaliana</i> in vitro	High for YbiA; different substrate than Schuller
PMID: 40929802 (Yang 2025)	RNAi + survival assay (ortholog)	Qualifies ND	Insect NADAR in immune defense	Beetle CmNADAR1/2 RNAi → reduced survival + intestinal necrosis during infection	<i>C. montrouzieri</i> ; in vivo RNAi	Medium; different species; no MF data
PMID: 20133860 (Estes 2010)	Expression / mutant phenotype	Supports ND	irg-1 as immune marker	irg-1::GFP reporter strongly induced by PA14; zip-2 required for induction	<i>C. elegans</i> intestine, <i>P. aeruginosa</i>	High; no MF data generated

Citation	Evidence Type	Direction	Claim Tested	Key Finding	Context	Confidence
PMID: 22520465 (Dunbar 2012)	Expression pattern	Supports ND	irg-1 regulation mechanism	zip-2/irg-1 pathway activated by pathogen-induced translational block	<i>C. elegans</i> intestine	High; no MF data generated
PMID: 38471365 (Xiao 2024)	Expression pattern	Supports ND	irg-1 as immune readout	SAM diet upregulates irg-1 via H3K4me3 pathway	<i>C. elegans</i> , PA14 infection	High; use irg-1 as reporter only
PMID: 33259473 (Vasquez-Rifo 2020)	Genetic / expression	Supports ND	zip-2 pathway context	PA14 cleaves ribosome H69; zip-2 pathway (including irg-1) antagonizes this	<i>C. elegans</i> intestine	High; pathway context, not MF
PMID: 22399070 (Aravind 2012)	Computational	Qualifies ND	NADAR acts on ADP-ribose derivatives	Predicted NADAR as NAD-utilizing pathway component; partially validated by later work	Bioinformatic analysis	Low-medium; prediction
PMID: 25027823 (Aravind 2014)	Review / evolutionary	Qualifies ND	NADAR in conflict/immunity	NADAR diversified in bacterial conflict systems; present in immunity-	Evolutionary analysis	Low; review-level

Citation	Evidence Type	Direction	Claim Tested	Key Finding	Context	Confidence
				related proteins		
PMID: 41505235	Direct assay (domain family)	Qualifies ND	NADAR hydrolase specificity	Synthetic ADP-ribosylated DNA substrates characterize hydrolase specificity	In vitro chemistry	High for chemistry indirect for irg-1
UniProt O16327	Database review	Supports ND	No catalytic activity annotated	No EC number; annotation score 2.0; PE=2	Curated database	High (curated absence)
WormBase WBGene00015574	Database phenotype	Supports ND	No loss-of-function phenotype	12 phenotypes NOT observed; 0 observed in standard assays	<i>C. elegans</i>	Medium (negative data)

GO Curation Implications

Current Annotation Status

- **GO:0003674 (molecular_function)** — ND evidence, GO_REF:0000015 — **ACCEPT is appropriate**
- **GO:0005575 (cellular_component)** — ND evidence, GO_REF:0000015
- **GO:0045087 (innate immune response)** — HEP,

P 16968778

- **GO:0050829 (defense response to Gram-negative bacterium)** — IEP,

P 20133860

- **GO:0140367 (antibacterial innate immune response)** — IEP,

P 20133860

Assessment

The ND annotation for molecular function should be **retained (ACCEPT)**. The evidence supports this because:

1. **No direct enzymatic assay** has been performed on *irg-1* protein
2. **No IEA annotation** for molecular function currently propagates from InterPro2GO (the NADAR IPR012816 entry would need to be checked for current GO term mappings)
3. The closest experimentally characterized homologs are in bacteria (*E. coli* YbiA) and oomycetes (*Phytophthora* NADAR), which are phylogenetically distant from *C. elegans*
4. *C. elegans* lacks riboflavin biosynthesis (the canonical NADAR substrate context for YbiA), raising questions about substrate specificity for this specific gene product
5. Functional heterogeneity within the NADAR family (riboflavin intermediates vs. guanine-ADP-ribose substrates) means domain presence alone is insufficient to predict precise substrate specificity

Future Annotation Candidates (Leads, Not Recommendations)

If future experimental work confirms *irg-1* enzymatic activity, candidate GO terms would include: - **GO:0016799** (hydrolase activity, hydrolyzing N-glycosyl compounds) — currently assigned to *E. coli* YbiA by IDA - A new or existing term for "guanine de-ADP-ribosylase activity" if one is created following the Schuller et al. 2023 findings - These should **NOT** be assigned to *irg-1* without direct experimental evidence or a robust ISS/IBA framework linking *irg-1* to a tested NADAR ortholog with conserved catalytic residues

Conflicts and Alternatives

No Direct Conflicts

The ACCEPT decision is not contested by any available evidence. No study claims to have identified *irg-1* molecular function. The hypothesis that the ND annotation is appropriate is universally consistent with the published literature.

Alternative Interpretations

- 1. NADAR as an immune effector with enzymatic activity:** The convergent use of NADAR domain proteins in immune defense in both nematodes (*irg-1*) and insects (ladybird beetles, [PMID: 40929802](#)) suggests the NADAR enzymatic activity may itself be the effector mechanism. If *irg-1* acts as a guanine de-ADP-ribosylase during infection, it could potentially: (a) reverse pathogen-induced ADP-ribosylation of host nucleic acids, (b) degrade pathogen-derived ADP-ribosylated substrates, or (c) modulate host signaling through ADP-ribose metabolism. None of these have been tested.
 - 2. NADAR as a non-enzymatic effector:** It is possible that *irg-1* has lost enzymatic activity and functions through protein-protein interactions or other non-catalytic mechanisms. The lack of observable phenotype upon gene disruption could be consistent with this.
 - 3. Paralog redundancy:** *C. elegans* has multiple NADAR domain-containing proteins. Any future ISS-based annotation must carefully confirm orthology rather than simply relying on domain presence. The lack of *irg-1* loss-of-function phenotype may reflect functional redundancy.
 - 4. Substrate context uncertainty:** Frelin et al. (2015) showed *E. coli* YbiA has N-glycosidase activity specific to riboflavin intermediates, while Schuller et al. (2023) showed a different subset of NADAR proteins hydrolyze ADP-ribose from guanine. This functional heterogeneity within the NADAR family means domain presence alone cannot predict substrate specificity for *irg-1*.
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Knowledge Gaps

Gap	What Was Checked	Why It Matters	Resolving Evidence
No enzymatic assay on irg-1	PubMed (25 papers), UniProt, WormBase, QuickGO	Cannot assign MF term without direct or strong ortholog evidence	Recombinant irg-1 protein assayed for guanine de-ADP-ribosylation and/or N-glycosidase activity
Catalytic residue conservation unknown	Domain-level data from Schuller 2023 (PDB 8BAU) and Frelin 2015	If catalytic residues are mutated, irg-1 may be a pseudoenzyme	Sequence alignment of irg-1 NADAR domain with catalytic site residues from characterized NADAR proteins
Unknown in vivo substrate	Literature on NADAR substrates; <i>C. elegans</i> metabolic context	<i>C. elegans</i> lacks riboflavin biosynthesis; unknown what irg-1 acts on	Substrate profiling or metabolomics during infection in irg-1 mutants
No loss-of-function phenotype	WormBase phenotype data (12 phenotypes NOT observed)	May indicate redundancy with other NADAR paralogs or assay insensitivity	Combinatorial RNAi/CRISPR of multiple NADAR genes during PA14 infection
Subcellular localization unknown	QuickGO (ND for cellular component)	Localization would constrain functional hypotheses	CRISPR-tagged endogenous irg-1::GFP; co-localization with organelle markers
Uncertain orthology among <i>C. elegans</i> NADAR paralogs	InterPro, UniProt	Multiple NADAR proteins in <i>C. elegans</i> ; unclear which is functional equivalent of characterized orthologs	Phylogenetic analysis of nematode NADAR family with functional paralogs mapped

Discriminating Tests

1. **Recombinant irg-1 enzymatic assay** (highest priority): Express and purify irg-1 protein. Test for: (a) guanine de-ADP-ribosylation activity using ADP-ribosylated guanine substrates (following Schuller et al. 2023 protocol and using synthetic substrates from [PMID: 41505235](#)), (b) N-glycosidase activity against various N-glycosidic substrates. This single experiment would directly resolve the MF annotation question.
 2. **Catalytic site sequence analysis** (immediate, computational): Align irg-1 NADAR domain with *Phytophthora* NADAR (PDB: 8BAU) and *E. coli* YbiA to assess whether catalytic residues are conserved. This can be done immediately and would strengthen or weaken the prediction.
 3. **irg-1 CRISPR knockout + PA14 infection survival** (medium priority): While standard assays showed no phenotype, a targeted slow-killing assay with PA14, combined with simultaneous knockout of other NADAR paralogs, may reveal an immune defense role analogous to insect NADAR knockdown.
 4. **AlphaFold structural comparison** (computational): Compare AlphaFold model of irg-1 (AF-O16327-F1) with the *Phytophthora* NADAR-ADP-ribose complex (PDB: 8BAU) to assess whether the ADP-ribose binding site is conserved.
 5. **Catalytic site mutagenesis** (if activity found): Mutate predicted catalytic residues and test whether irg-1 immune function requires enzymatic activity, distinguishing between enzymatic and non-catalytic effector roles.
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Curation Leads

All leads below require curator verification.

Lead 1: ACCEPT the ND annotation (Primary Recommendation)

- **Action:** ACCEPT (no change to current annotation)
- **Confidence:** High
- **Rationale:** No experimental evidence exists for irg-1 molecular function. The ND annotation correctly reflects this state.

Lead 2: Update the review rationale text with recent citations

- **Suggested change:** The current rationale mentions "NADAR domain... associated with ADP-ribosylhydrolase activity, particularly the ability to remove ADP-ribose from guanine bases." This is now directly supported by Schuller et al. 2023 ([PMID: 37390817](#)), which demonstrated guanine de-ADP-ribosylation by eukaryotic NADAR proteins. The rationale should cite this paper to strengthen the contextual justification while maintaining the ND conclusion.
- **Candidate reference:**

 37390817

(Schuller et al. 2023, Mol Cell)

- **Candidate snippet to verify:** "We show that guanine de-ADP-ribosylation is also conserved among eukaryotic and non-DarT-associated NADAR members, indicating a wide distribution of reversible guanine modifications beyond DarTG systems."

Lead 3: Monitor for future ISS/IBA annotation opportunity

- **Context:** If an experimentally characterized NADAR ortholog with confirmed guanine de-ADP-ribosylase activity is established as a clear 1:1 ortholog of *irg-1*, an ISS annotation to GO:0016799 (hydrolase activity, hydrolyzing N-glycosyl compounds) or a more specific child term could be warranted.
- **Blocking issue:** The presence of multiple NADAR domain proteins in *C. elegans* complicates orthology assignment. Clear phylogenetic evidence would be needed.

Lead 4: Consider adding

 40929802

to supporting references

- **Context:** The 2025 ladybird beetle NADAR paper (Yang et al.) provides cross-phylum evidence that NADAR domain proteins function in intestinal immune defense, strengthening the biological plausibility that *irg-1* is an immune effector. This is supporting context for biological process annotations, not the molecular function annotation under review.

Lead 5: Flag irg-1 as a priority target for experimental characterization

- **Context:** While irg-1 is overwhelmingly used as a transcriptional reporter in the literature, the NADAR domain biochemistry and cross-species functional data suggest it likely has an active molecular function. The protein is small (212 aa), expressed in a known tissue (intestine), and amenable to recombinant expression. The ND annotation should be maintained but the gene flagged for re-evaluation as the NADAR field advances rapidly (three major papers in 2023–2025).
- **Suggested curator question:** "Has any group published or deposited data on irg-1 protein purification, enzymatic assay, or structural characterization since mid-2025?"

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